

Roadmap to Master AI Agents (Industrial Level)

Mathematics & Theory

Linear Algebra Vectors, matrices for ML models

Calculus Gradient Descent, optimization

Probability & Statistics Predictions, distributions

Discrete Math Logic, state transitions

Information Theory For decisions & communication

Programming

Python NumPy, Pandas, Scikit-learn, Matplotlib

Bash/Linux Automation & scripting

C++ Real-time agents, robotics (optional)

JavaScript For web-based agent UIs (optional)

Machine Learning

Supervised/Unsupervised Learning scikit-learn, XGBoost

Model Evaluation Precision, Recall, AUC

Deep Learning TensorFlow, PyTorch

Model Deployment Flask, FastAPI, Docker

Reinforcement Learning

MDPs, Q-Learning, SARSA

Deep Q Networks (DQN), PPO, A3C

Libraries PyTorch, Stable-Baselines3, RLlib

Environments OpenAI Gym, Unity ML Agents

Large Language Models & Generative AI

Prompt Engineering OpenAI, Anthropic, Cohere

LangChain Building LLM agents

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Vector Databases FAISS, Pinecone, ChromaDB

Embeddings & RAG Smart retrieval with memory

Software Engineering

Version Control Git, GitHub

Design Patterns Scalable architecture

APIs REST, FastAPI

Testing & Debugging Logs, Unit tests

Tool Integration

LangChain, AutoGen, AgentOps LLM Agents

Selenium / Playwright Browser automation

Zapier / APIs Real-world integrations

Docker Containers

Redis / PostgreSQL / MongoDB Data management

Cloud & Deployment

AWS / GCP / Azure Cloud services

CI/CD Jenkins, GitHub Actions

Docker & Kubernetes Scalability

Monitoring & Logging Tracking performance

Domain Knowledge

Finance Trading bots, fraud detection

Healthcare Diagnosis agents

Robotics Navigation & planning agents

NLP Chatbots, document agents

Gaming AI NPCs, decision agents

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Soft Skills & Practices

Problem Solving Real-world thinking

Communication Explaining models

Collaboration Teamwork

Research Reading Stay updated (Arxiv, GitHub)